

Modeling perspectives on the intestinal crypt, a canonical system for growth, mechanics, and remodeling

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Abstract

The intestinal epithelium is crucial to maintaining a healthy gut. Central to this are the crypts of Lieberkühn, which coordinate the rapid self-renewal of the epithelium to protect the small intestine and colon during digestion. Further, mutations in crypt cells may initiate colorectal cancer. It is, therefore, important to understand how crypts function during homeostasis and disease. Mathematical and computational modeling has contributed to increasing our understanding of crypt dynamics. However, many open questions remain to be addressed, particularly regarding the role of mechanics in intestinal crypt dynamics. In this article, we review the state-of-the-art in crypt modeling and explain why further progress requires the integration of new theory from continuum mechanics with cell-based computational models and experimental data.

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Current Opinion in Biomedical Engineering 2020, 15:32–39

This review comes from a themed issue on **Biomechanics and Mechanobiology: Growth and remodeling in both mechanics and mechanobiology**

Edited by Ellen Kuhl and Jay Humphrey

For a complete overview see the [Issue](#) and the [Editorial](#)

<https://doi.org/10.1016/j.cobme.2019.12.012>

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Keywords

Intestinal crypt, Continuum mechanics, Cell-based computational models.

Introduction

The intestinal epithelium is a monolayer that protects the small intestine and colon during digestion and simultaneously facilitates nutrient absorption [44,19]. Its rapid self-renewal is controlled by millions of crypts of Lieberkühn, uniformly-spaced invaginations lining the gut. In the intestines, multipotent stem cells are localized in crypt bases where they coordinate cell proliferation, death, and migration to facilitate the constant turnover of the epithelium every 3–5 days [19], with increased turnover in chronic inflammation [40]. Consequently, the crypt has become a canonical model

for studying stem cell dynamics [5]. In general, proliferative cells are only found in the crypts. Hence, when gut tissue is injured or inflamed, crypts are crucial to restoring barrier function by generating new tissue [27]. Moreover, colonic cancer is known to originate in colonic crypts [5]. For these reasons, understanding how crypt dynamics maintain homeostasis in health, how they are restored following injury, and disrupted in disease is of high clinical importance.

While considerable research attention has been directed toward the crypt, many questions remain unresolved. For example, what processes ensure that, during homeostasis, each crypt maintains a uniform and robust structure, while large numbers of cells proliferate and migrate? Additionally, little is known about how the crypt's specialized cell populations are generated and how its invaginated structure arises during development. Such fundamental questions are intimately related to the concept of tissue growth and its interactions with tissue mechanics.

The crypt's compartmentalized nature renders it ideal for mathematical and computational modeling. Significant efforts have focussed on using cell-based approaches to model the crypt, validate existing hypotheses, and generate new ones [17]. While such theory has yielded insight into the dynamics of healthy and diseased crypts, more can be achieved, particularly regarding the role of mechanics in regulating subcellular signaling pathways and thereby rates of cell proliferation, migration, and death. This is now possible due to the development of morphoelasticity theory, which extends continuum mechanics to account for biological tissue growth [21].

This review provides an introduction to the crypt, and relevant modeling approaches. We describe the crypt structure and the signaling pathways known to contribute to homeostasis. Relevant modeling approaches (discrete and continuum) are then introduced, with their strengths and limitations, and a summary of the advances they have enabled. We close by discussing open questions that will advance the modeling field, and in turn, increase understanding of the role of biomechanics in regulating healthy and diseased intestinal crypts.

Biological background

We now introduce the main components of the crypt. We describe its structure and surrounding environment, including signaling pathways involved in its regulation of health and disease; the reviews by Gehart and Clevers [19] and Spit et al. [44] contain further details.

Each test-tube-shaped crypt is lined with an epithelial sheet (Figure 1). Adjacent colonic crypts are separated by flat tissue regions. In the small intestine, crypts are connected to large protrusions called villi that increase gut surface area to aid nutrient absorption. Each crypt contains a pool of slowly-cycling [45] (or even paused [10]) stem cells in its base. In the small intestine, Paneth cells are also found in the crypt base, where they secrete factors that support the stem cell niche (the stem cell pool and its microenvironment) [19,44]. As stem cells divide, their offspring push upward toward the lumen, in a conveyor belt fashion. Transit-amplifying cells proliferate rapidly, before differentiating into specialized, non-proliferating cells, such as goblet cells [44]. Directly underneath the crypt is an extracellular matrix network called the basement membrane, which aids crypt maintenance and facilitates

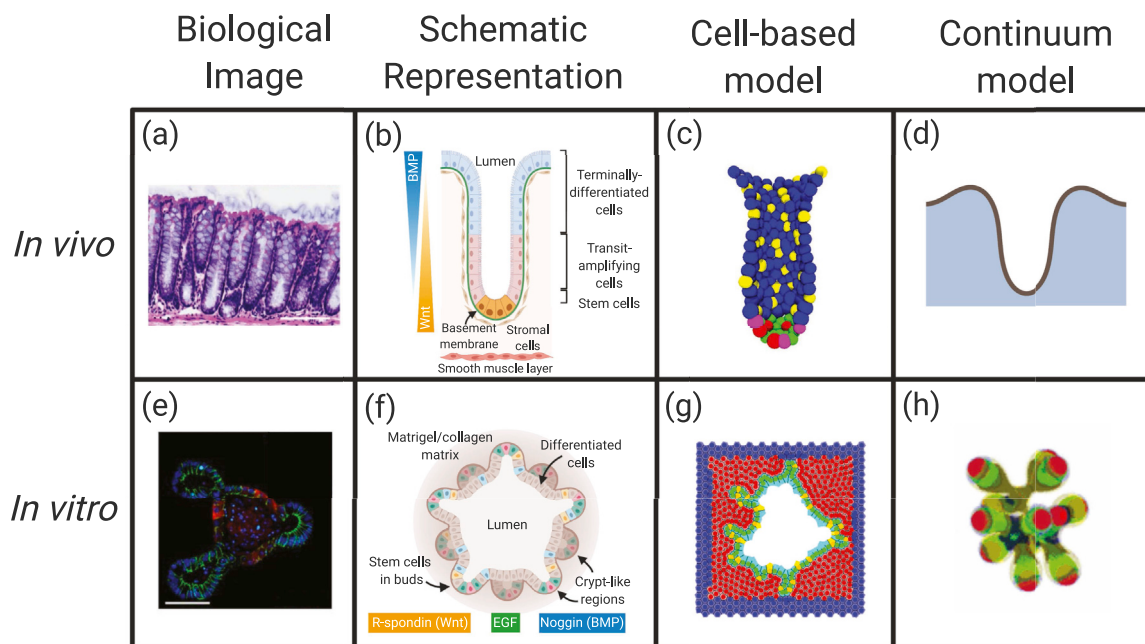
communication with stromal cells [34]. The stroma provides structural stability and additional regulatory signals [29]. It is supported by a layer of smooth muscle cells. A schematic of a crypt is presented in Figure 1(b).

The development of crypt organoids has accelerated understanding of crypt biology. Crypt organoids are *in vitro* cell cultures generated from single stem cells or isolated crypts, which grow into a confluent layer of epithelial tissue, with crypt-like buds surrounding an inner lumen (see Figure 1(f)). Organoids enable detailed investigation of cell–cell signaling, by knocking out genes and/or embedding them in different media. Moreover, as they are *in vitro* cultures, cell behavior can be tracked at greater spatiotemporal resolution using imaging techniques [20,42].

While multiple signaling pathways contribute to crypt homeostasis [19,44], we describe the dominant ones:

1. Wnt signaling is the principal signaling pathway in the crypt, and multiple Wnt proteins regulate cell proliferation, stemness, cell migration, differentiation, and apoptosis [19,44]. A decreasing Wnt

Figure 1



The crypt at varying levels of biological and mathematical abstractions. (a) Histological image of *in vivo* murine colonic crypts; the scale bar represents 100 μm (adapted with permission from Shoshkes-Carmel et al. [43]). (b) Schematic diagram of a single colonic crypt and essential components. (c) 3D crypt model by Thalheim et al. [48] (adapted with permission). (d) 2D continuum representation of crypt (adapted with permission from Almet et al. [1]). (e) Fluorescence image of *in vitro* murine small intestinal organoid; the scale bar represents 50 μm (adapted with permission from Gjorevski et al. [20]). (f) Schematic diagram of the organoid structure and relevant components. (g) 2D organoid model used in Langlands et al. [31] (adapted with permission). (h) 3D colon cancer organoid model from Yan et al. [51] (reproduced with permission).

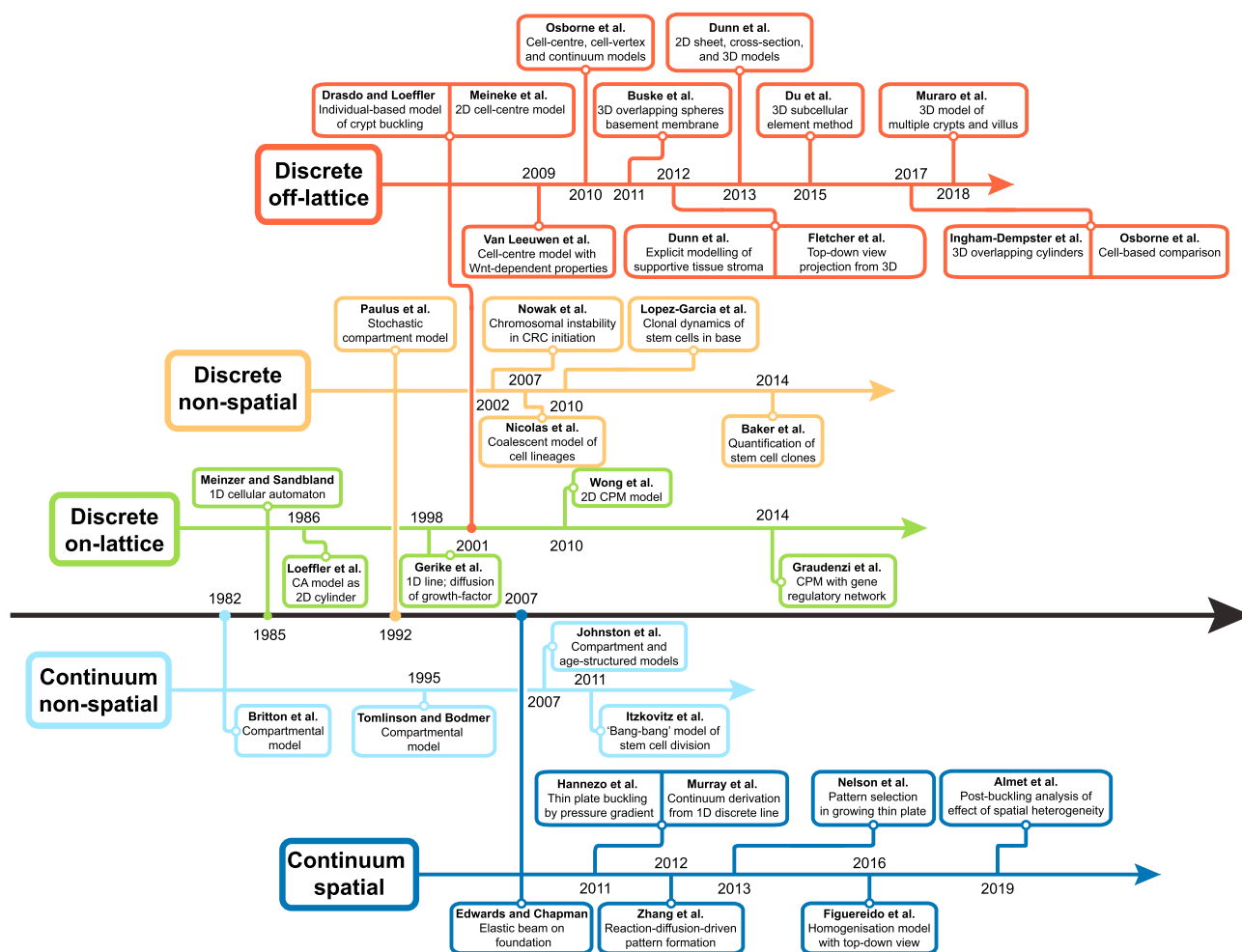
gradient exists along the crypt axis, but it is not fully known how this gradient arises or is maintained. Furthermore, Wnt pathway mutations are among the earliest to occur in colorectal cancer [19].

2. Bone Morphogenetic Protein (BMP) signaling suppresses stem cell multipotency and drives terminal differentiation. An increasing BMP gradient from the crypt base is maintained by the surrounding stroma, and BMP inhibition has been shown to cause excessive crypt formation [44].
3. Notch signaling governs cell fate specification [28]. In the small intestine, Notch-active stem and Paneth cells arise through lateral inhibition caused by cell–cell contact [44].
4. The Hippo pathway regulates Wnt and Notch signaling [4,42]. Inactive during homeostasis, its transducers YAP and TAZ facilitate regeneration

following injury [47], while their loss results in uncontrolled proliferation [23].

The involvement of YAP and TAZ in crypt homeostasis is significant because of their mechanotransductive roles; their regulation of Wnt and Notch is triggered by mechanical cues. Indeed, Gjorevski et al. [20] showed that culturing organoids in soft matrices increased stem cell proliferation, while culturing them in stiffer matrices stimulated stem cell differentiation and organoid formation, through a YAP-dependent pathway. Serra et al. [42] showed that transient activation of YAP/TAZ was required to trigger Paneth cell differentiation, which introduced mechanical heterogeneity in organoids, leading to crypt budding. Despite increasing evidence suggesting a regulatory role for mechanical

Figure 2



A timeline of modeling approaches that have been applied to the crypt. Models have been highlighted for the novelty of their approach and classified according to the discrete or continuum approach that is used, and the spatial effects that are considered.

stimuli, such mechanisms can be difficult to probe experimentally. In such situations, mathematical and computational modeling can provide significant insight.

Mathematical and computational models of the crypt

The approaches reviewed here can be termed “discrete”, where the crypt is viewed as a collection of individual cells (see Figure 1(c and g)) or “continuum”, where it is modeled as a smooth tissue (Figure 1(d and h)). Here, we focus on spatially-resolved models (for a review of models that neglect spatial effects, see Carulli et al. [11]). We discuss their strengths and weaknesses, focusing on recent modeling studies (see Figure 2 for a timeline of crypt modeling approaches).

Cell-based models

Cell-based models represent each cell as one or several points in space with a prescribed cell region [35]. They permit extraction of cell-scale data, such as migration velocities and lineage [48,39], and tissue scale information, such as mitotic distributions [13], for comparison with experimental data [5]. An example of a recent and comprehensive cell-based model of the crypt is that of Thalheim et al. [48]. We use this model as a benchmark for comparison with other models.

Thalheim et al. consider a 3D overlapping spheres model, in which crypt geometry is specified by a surface of revolution (see Figure 1(c)), a common representation among 3D models [13,12,36]. In 2D geometries, cell shape can be modeled via a Voronoi Tessellation, with spatial connectivity defined through a Delaunay Triangulation [31,2,39]; this is avoided in 3D due to computational costs. Adhesion to the crypt surface is enforced by a triangular fiber network, representing the basement membrane and included to ensure shape stability [26,31,2]. Cells interact with the membrane through a weakly-adhesive force. Detachment from the basement membrane results in cell death [13,31,2]. Cell-based models typically include forces that restrict cells to the crypt surface [13,12,36]. Cells at the crypt top are removed from simulations [12,13,39]. In Muraro et al. [36], cells migrate out of the crypt and onto the villus.

Thalheim et al. model cells as elastic spheres, using Langevin dynamics to update cell positions and sizes. In other words, cells were modeled as cylinders [26], while the subcellular element method represented cells as collections of overlapping spheres [12]. Langevin equations incorporate adhesion forces between overlapping cells, deformations induced by cell–cell contact, changes to cell volumes, and interactions with the basement membrane. A popular alternative approach to update cell positions is an over-damped form of Newton’s Second Law [31,2,36,13,39,26]. In Thalheim et al., proliferating cells grow in stochastic increments,

dividing into two cells when their initial volume doubles [12]. If cells are mechanically compressed, proliferation may stop due to contact inhibition [13,39]. Cell proliferative capacity and fate specification are determined by Wnt and Notch levels. Wnt signaling is modeled as a decreasing gradient from the base, a *de facto* approach [31,2,36,13,39,26,12]. Stem cells that leave the base to become transit-amplifying cells, which differentiate as they migrate toward the crypt lumen. Notch activity is determined by Notch levels from the nearest neighbors. Stem cells are assumed to express high levels of Notch. Transit cells differentiate into either Paneth cells (high Wnt, low Notch) that migrate down toward the base, or Goblet cells (low Wnt, low Notch) that migrate up the crypt. Du et al. [12] consider the additional role of BMP signaling in preventing the over-expansion of the stem cell population.

A strength of cell-based models is their ability to integrate processes acting on different spatial and temporal scales within a multiscale framework. For example, logic-based rules at the cell scale may be coupled to subcellular mathematical models for gene regulatory networks [22], which model gene activation as binary states, or ordinary differential equations for biochemical signaling pathways that describe continuous changes [50]. This allows more biological detail to be incorporated and investigation of how perturbations subcellular processes, for example due to genetic mutations, may impact tissue-level behavior.

However, with greater biological detail comes greater model complexity. For cell-based models, there is no standard way to analyze simulation results, which are obtained by averaging multiple realizations. Furthermore, incorporating biological detail increases the number of model parameters. As the dimension of the parameter space increases, it becomes more difficult to identify the mechanism(s) driving observed results and/or to fit the models to experimental data. Individual-based models can be computationally expensive to simulate, particularly when 3D geometries and/or sophisticated implementations, such as the immersed boundary method, which accounts for fluid–structure interactions between cells [16], are used.

Cell-based models also generally assume a fixed geometry. Processes, such as crypt invagination [46] and fission [31] occur in development, cancer, and regeneration, but cannot be modeled properly without accounting for deformability. As forces are specified at the cell scale, it can be difficult to estimate deformations on the scale observed *in vivo* [13,48]. The rigid geometry, thus, limits the scope of what can be investigated. While some cell-based organoid models account for tissue deformation [41,31,2,49], they do not produce deformations on the scale needed to study fission or invagination.

Continuum models

When studying processes characterized by large deformations, such as crypt fission or invagination, an alternative approach is to focus on the macroscale. The crypt tissue may be represented as multiple, interacting diffusible species [15,51], viscous fluids [30], or elastic sheets [25,18]. Morphoelasticity theory, which extends nonlinear elasticity to account for tissue growth [21], is another promising approach to tackle problems that cannot be addressed with cell-based models.

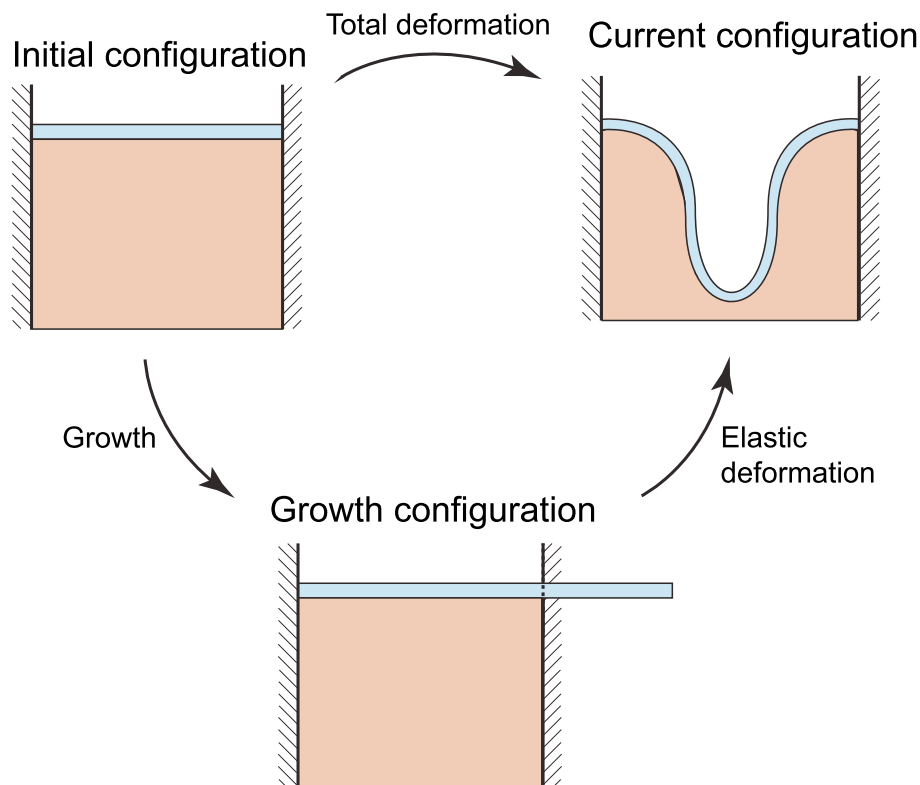
Continuum mechanics models are rooted in more developed branches of mathematics than cell-based models; governing equations come from first principles and deformations arise naturally due to mechanical forces and torques. As growth typically occurs on longer timescales than those associated with elastic/viscoelastic deformations, quasi-static mechanical equilibrium may be assumed. At each time step, the material deformation gradient can be decomposed into distinct growth and elastic deformations. First, growth is defined locally, and different tissue regions may overlap after growth. The elastic deformation map then reassembles the grown material to be geometrically compatible, which may induce residual stresses (see Figure 3).

Morphoelasticity has been used to study many biological processes [3], including wound healing, artery remodeling, and brain cortical folding, the latter sharing many similarities with crypt invagination. Several continuum-mechanics models for crypt and/or villi buckling have been proposed [14,37,8,38], but lack biological specificity. Morphoelasticity has recently been used to study cylindrical buckling along the gastrointestinal tract [7], while Almet et al. [1] considered the effect of material heterogeneity and large deformations on the growth of an idealized crypt model (Figure 1(d)). Despite these advances, the biological realism of continuum crypt models remains limited.

Continuum frameworks, formulated using ordinary and partial differential equations, can be analyzed using a range of analytical tools, including asymptotic methods and linear stability and weakly nonlinear analysis. In this way, it is possible to assess the qualitative and quantitative effects of varying different model parameters and systematically reveal the mechanisms driving observed behaviors.

There are several challenges with tissue-level continuum modeling. By construction, in such models, there is

Figure 3



Potential application of morphoelasticity to the crypt. Here, the crypt is treated as a growing, elastic rod. Crypt evolution is decomposed into an axial growth phase, as a proxy for tissue proliferation, and the subsequent elastic deformations due to residual stresses induced by growth, mechanical effects of the underlying substrate, and boundary conditions.

Table 1

Desirable features of an ideal crypt model, and the current efficacy of continuum and cell-based frameworks to capture them. The two classes of frameworks have complementary strengths and limitations.

Crypt feature	Continuum	Discrete (cell-based)
Deformability	✓✓: Can use continuum mechanics theory	✓: Not well-developed
Mechanical effects	✓✓: Very well-placed	
Clonal cell tracking	✓: Possible for a few models	✓✓: Can easily track cell lineages
Subcellular detail	Has not been considered in detail	✓✓: Very well-suited for this
Stochasticity	×	✓
Cell migration	Possible for certain models	✓✓
Population control	✓: Laws to prescribe local growth	✓✓
Interaction with external environment	✓✓: Can prescribe mechanical forces	✓: Computationally expensive
Cellular heterogeneity	✓: Coarse-grained heterogeneity	✓✓: Individual cell heterogeneity
Cell sorting	×	✓: Possible in some models
Experimental parametrization	Possible but has not been implemented	✓: Possible with enough biological detail

no individual cellular unit, despite the need to incorporate relevant cellular-level detail. This is evident in the crypt, which is spatially and temporally heterogeneous in numerous ways, including growth capacity and stiffness, which depend on cellular processes. Describing such features in a continuous setting is non-trivial, but feasible. For instance, one may need to couple the various processes, spanning subcellular, cellular, and tissue scales. However, multiscale approaches, such as homogenization theory or micro-mechanics, are, in principle, available. Parameterizing such models poses a different challenge. For instance, a key mechanical parameter is the stiffness, e.g. Young's modulus; stiffness is easily obtained with a macroscale material, but less so with a small biological sample. Atomic Force Microscopy has become useful for measuring the mechanical properties of individual cells, including stem and Paneth cell stiffnesses [41]. However, there is a clear need to perform such experiments across the entire crypt, although it is unclear how the stiffness that may be determined from a biological experiment is related to the value in the continuum setting.

Outlook

As crypt modeling has become more mature and experimental data collection methods more sophisticated [24], our understanding of the crypt and its role in gut health has increased. The next challenge is to understand how biochemical and biomechanical processes interact across multiple spatiotemporal scales. A particular challenge relates to crypt fission: during fission, a bifurcation at the crypt base results in the formation of two new crypts. In disease, crypt fission drives the expansion of adenomas, allowing mutated crypts to spread [5,31], but it is also vital for tissue

regeneration after injury [23]. Further complicating the issue, it has been recently suggested that crypts may fuse together [9,6]. The ubiquity of crypt fission, during homeostasis, cancer and regeneration, make understanding how it arises of great clinical significance, while its distinctive mechanical features, particularly the tissue deformations akin to buckling, render it amenable to investigation with mechanistic modeling.

Addressing the above challenges requires further development of continuum and cell-based models, particularly biologically-specialized continuum models of the crypt. We believe that morphoelasticity theory is well-suited to fill this void, but doing so in a way that is consistent with cell activity is a significant challenge. A related challenge is model validation against experimental data. As it is now possible to collect dynamic, cell-scale data [24], this challenge can start to be addressed.

As indicated in Table 1, neither cell-based nor continuum frameworks capture all aspects of an "ideal" crypt model. Instead, we advocate an integrated approach: by linking discrete and continuum models by, say, mapping model parameters, we can harness the complementary advantages of each approach and switch between them, depending on the considered problem. These studies are also needed to establish the robustness of model predictions to changes in the assumptions about microscopic processes.

Another significant challenge concerns parameter estimation. Aside from the sheer number of parameters and uncertainty in their values, equally important for a given model is the number of chosen rules that do not emerge from first principles. In evaluating the results of a model,

the ultimate test is whether it can reproduce data. But even though research is becoming more interdisciplinary [36], many models are not validated against biological data. As experimental data become more detailed and widely available, and as models become more complex and multiscale, in attempting to connect the two, it becomes ever more important to consider how parameter uncertainty affects results. Model inference tools represent a promising way forward [32,33], but significant work remains to be done.

Conflict of interest statement

Nothing declared.

Acknowledgments

This work was supported by Cancer Research UK (CRUK) grant number C5255/A23225, through a Cancer Research UK Oxford Centre Prize DPhil Studentship.

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